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ASSIGNMENT 2

BEVEL GEAR DESIGN

● Given:

- Power: 10 KW
- $N_1 = 1440$ RPM
- $N_2 = 350$ RPM

➤ PROGRAM:

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BEVEL GEAR.py - C:\Users\kaziv\OneDrive\Desktop\DESIGN OF MACHINE SYSTEMS\ASSIGNMENT 2 BEVEL GEAR PROGRAM\BEVEL GEAR.py (3.12.6)
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# DMS ASSIGNMENT 2 BEVEL GEAR
# STEP 0: Importing Relevant modules
import math
from tabulate import tabulate

# STEP 1: Select type of system.
print("-----")
print("Select the type of system:")
print("1. Closed System")
print("2. Open System")
SYSTEM_TYPE = input("Select type of system: ")
# STEP 2: Select type of teeth profile.
print("Select type of teeth profile:")
print("1. Involute teeth profile")
print("2. Cycloidal teeth profile")
TEETH_TYPE = input("Select type of teeth profile: ")
# STEP 3: Assigning Pressure Angle and Angle at which the shafts intersect.
PRESSURE_ANGLE = int(input("Enter the pressure angle: "))
INTERSECTION_ANGLE = int(input("Enter the angle at which the shafts intersect: "))
# STEP 4: LAYOUT
# STEP 5: Determination of initial gear ratio.
INITIAL_SPEED = int(input("Initial speed in RPM: "))
FINAL_SPEED = int(input("Final speed in RPM: "))
INITIAL_GEAR_RATIO = INITIAL_SPEED / FINAL_SPEED
# STEP 6: Determination of cone angles.
COT_DELTA_1 = (INITIAL_GEAR_RATIO + math.cos(math.radians(INTERSECTION_ANGLE)))/math.sin(math.radians(INTERSECTION_ANGLE))
DELTA_1 = math.degrees(math.atan(1/COT_DELTA_1))
DELTA_2 = INTERSECTION_ANGLE - DELTA_1
# STEP 7: Number of teeth in gear and pinion.
PINION_TEETH = 18
GEAR_TEETH = math.ceil(((PINION_TEETH*INITIAL_GEAR_RATIO)+1))
GEAR_RATIO = GEAR_TEETH/PINION_TEETH
# STEP 8: Virtual number of teeth in gear and pinion.
PINION_VIRTUAL_TEETH = math.ceil(PINION_TEETH/math.cos(math.radians(DELTA_1)))
GEAR_VIRTUAL_TEETH = math.ceil(GEAR_TEETH/math.cos(math.radians(DELTA_2)))
# STEP 9: Material Selection.
SIGMA_U_PINION = 720
SIGMA_Y_PINION = 380
SIGMA_U_GEAR = 570
SIGMA_Y_GEAR = 310
BHN_PINION = 241
BHN_GEAR = 187
# STEP 10: Lewis form factor.
Yv1 = (0.154-(0.912/PINION_VIRTUAL_TEETH))*3.14
Yv2 = (0.154-(0.912/GEAR_VIRTUAL_TEETH))*3.14
# STEP 11: Determining if the pinion is the weakest element.
SIGMA_Y1 = SIGMA_Y_PINION*Yv1
SIGMA_Y2 = SIGMA_Y_GEAR*Yv2

# STEP 12: Determining the design torque.
MOTOR_POWER = input("Enter the power: ")
Mt = ((1.3*int(MOTOR_POWER)*1000*60)/(2*3.14*int(INITIAL_SPEED)))*1000
# STEP 13: Design bending stress.
SIGMA_B = ((1.4/3)*((0.35*SIGMA_U_PINION*10)+1200))*0.1
# STEP 14: Module.
Si = 10
Module = math.ceil(((Mt/(Yv1*SIGMA_B*Si*PINION_VIRTUAL_TEETH))**(1/3))*1.28*(6/5)*1.3)
# STEP 15: Checking wear.
DESIGN_COMPRESSIVE_STRESS = BHN_PINION*25*1*0.1
R = 0.5*Module*(((PINION_TEETH**2)+(GEAR_TEETH**2))**(1/2))
B = R/3
E = 215000
INDUCED_COMPRESSIVE_STRESS = (0.72/(R-(0.5*B)))*((E*Mt*((1+(GEAR_RATIO**2))**3)**(1/2))/(GEAR_RATIO*B))**(1/2))

# STEP 15: Results and conclusion.
print("-----")
print("-----BEVEL GEAR DESIGN-----")
print("-----")
Results = [{"Type of system selected is: ", SYSTEM_TYPE, "Used for industrial purposes."},
[2, "Type of teeth profile selected is: ", TEETH_TYPE, "Easy to manufacture."],
[3, "Pressure angle is: ", PRESSURE_ANGLE, "-"],
[4, "Angle at which the shafts intersect each other is: ", INTERSECTION_ANGLE, "-"],
[5, "Initial speed is: ", INITIAL_SPEED, " RPM"],
[6, "Final speed is: ", FINAL_SPEED, " RPM"],
[7, "Delta 1: ", round(DELTA_1, 2), "Degrees."],
[8, "Delta 2: ", round(DELTA_2, 2), "Degrees."],
[9, "Number of teeth on pinion: ", PINION_TEETH, "Minimum number of teeth."],
[10, "Number of teeth on gear: ", GEAR_TEETH, "Hunting teeth added."],
[11, "Gear ratio: ", round(GEAR_RATIO, 2), "-"],
[12, "Virtual number of teeth on pinion: ", PINION_VIRTUAL_TEETH, "-"],
[13, "Virtual number of teeth on gear: ", GEAR_VIRTUAL_TEETH, "-"],
[14, "Pinion material: ", "C-50", "-"],
[15, "Ultimate stress for pinion: ", round(SIGMA_U_PINION, 2), " N/mm2"],
[16, "Yield stress for pinion: ", round(SIGMA_Y_PINION, 2), " N/mm2"],
[17, "Gear material: ", "C-35", "-"],
[18, "Ultimate stress for gear: ", round(SIGMA_U_GEAR, 2), " N/mm2"],
[19, "Yield stress for gear: ", round(SIGMA_Y_GEAR, 2), " N/mm2"],
[20, "Lewis form factor for pinion: ", round(Yv1, 2), "-"],
[21, "Lewis form factor for gear: ", round(Yv2, 2), "-"],
[22, "Input motor power: ", MOTOR_POWER, " KW"],
[23, "Design bending stress is: ", round(SIGMA_B, 2), " N/mm2"],
[24, "Design Motor torque is: ", round(Mt, 2), " N/mm2"],
[25, "Module: ", round(Module, 2), " mm"],
[26, "Cone Distance is: ", round(R, 2), " mm"],
[27, "Face width is: ", round(B, 2), " mm"],
[28, "Design compressive stress is: ", round(DESIGN_COMPRESSIVE_STRESS, 2), " N/mm2"],
[29, "Induced compressive stress is: ", round(INDUCED_COMPRESSIVE_STRESS, 2), " N/mm2"]]
print(tabulate(Results, headers=["SrNo", "PARAMETERS", "VALUES", "UNITS / REMARKS"]))
print("-----")
print("Conclusion: ")
if SIGMA_Y1 < SIGMA_Y2:
    print("Pinion is chosen / kept as a weaker element because it is easy to replace it incase of failure, Therefore its design is done.")
else:
    print("Change the material, pinion is taken as the weaker material as it can be easily changed")
if DESIGN_COMPRESSIVE_STRESS > INDUCED_COMPRESSIVE_STRESS:
    print("Design is Safe since induced compressive stress (", round(INDUCED_COMPRESSIVE_STRESS, 2),
    " N/mm2")
    " is less than the design compressive stress. (", round(DESIGN_COMPRESSIVE_STRESS, 2), " N/mm2")
    ")")
else:
    print("Design isnt safe, increase the modulus or hardness.")
print("-----")

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➤ USER INPUT:

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= RESTART: C:\Users\kaziw\OneDrive\Desktop\DESIGN OF MACHINE SYSTEMS\ASSIGNMENT 2 BEVEL GEAR PROGRAM\BEVEL GEAR.py
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Select the type of system:
1. Closed System
2. Open System
Select type of system: Closed System.
Select type of teeth profile:
1. Involute teeth profile
2. Cycloidal teeth profile
Select type of teeth profile: Involute teeth profile.
Enter the pressure angle: 20
Enter the angle at which the shafts intersect: 85
Initial speed in RPM: 1440
Final speed in RPM: 350
Enter the power: 10
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➤ OUTPUT:

-----BEVEL GEAR DESIGN-----			
SrNo	PARAMETERS	VALUES	UNITS / REMARKS
1	Type of system selected is:	Closed System.	Used for industrial purposes.
2	Type of teeth profile selected is:	Involute teeth profile.	Easy to manufacture.
3	Pressure angle is:	20	-
4	Angle at which the shafts intersect each other is:	85	-
5	Initial speed is:	1440	RPM
6	Final speed is:	350	RPM
7	Delta 1:	13.34	Degrees.
8	Delta 2:	71.66	Degrees.
9	Number of teeth on pinion:	18	Minimum number of theeth.
10	Number of teeth on gear:	76	Hunting teeth added.
11	Gear ratio:	4.22	-
12	Virtual number of teeth on pinion:	19	-
13	Virtual number of teeth on gear:	242	-
14	Pinion material:	C-50	-
15	Ultimate stress for pinion:	720	N/mm2
16	Yield stress for pinion:	380	N/mm2
17	Gear material:	C-35	-
18	Ultimate stress for gear:	570	N/mm2
19	Yield stress for gear:	310	N/mm2
20	Lewis form factor for pinion:	0.33	-
21	Lewis form factor for gear:	0.47	-
22	Input motor power:	10	KW
23	Design bending stress is:	173.6	N/mm2
24	Design Motor torque is:	86252.65	N/mm2
25	Module:	4	mm
26	Cone Distance is:	156.2	mm
27	Face width is:	52.07	mm
28	Design compressive stress is:	602.5	N/mm2
29	Induced compressive stress is:	459.15	N/mm2

Conclusion:			
Pinion is chosen / kept as a weaker element because it is easy to replace it incase of failure, Therefore its design is done.			
Design is Safe since induced compressive stress (459.15) is less than the design compressive stress.(602.5)			
